

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

June Examination – Paper 1

5 June 2026

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Mrs S Rowell

Moderator

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INSTRUCTIONS

Read the following instructions carefully before answering the questions.

1. This question paper consists of 12 questions, 10 pages and a formula sheet. Please ensure that your question paper is complete. You may detach the formula sheet for your own reference.
2. Answer ALL the questions in the answer book provided. Please note clearly where the additional space is being used.
3. Clearly show ALL calculations, diagrams, graphs, etc. that you have used in determining your answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
6. If necessary, round off answers to TWO decimal places, unless stated otherwise.
7. Diagrams are NOT necessarily drawn to scale.
8. Use the mark allocation to guide the length of your solution.
9. Write neatly and legibly.

QUESTION 1

1.1 Solve for x :

1.1.1 $x(2x - 7) = -4$ (round off to TWO decimal places) (3)

1.1.2 $3^{2x-1} + 9^{x-1} - 4 = 0$ (5)

1.1.3 $\sqrt{2x+4} + 5 = x + 3$ (5)

1.1.4 $2x^2 - 11x \leq 6$ (3)

1.2 Solve for x and y simultaneously:

$$\begin{aligned} 2x - y - 1 &= 0 \\ x^2 - xy &= y^2 - 11 \end{aligned} \quad (6)$$

1.3 Consider the expression: $A(x) = \sqrt{x^2 - 8x + 20}$

1.3.1 Show that $A(x)$ is real for all real values of x . (2)

1.3.2 What is the minimum value of $A(x)$? (2)

[26]

QUESTION 2

A quadratic sequence is defined by $T_n = 2n^2 + bn + c$. The first term is -3 and the second term is 2.

2.1 Determine the constant second difference. (1)

2.2 Determine the values of b and c . (4)

[5]

QUESTION 3

3.1 Consider the geometric sequence: $x^{-3}; x^{-4}; x^{-5}; \dots$

3.1.1 Determine the general term of the sequence as a single base of x . (2)

3.1.2 Calculate T_{10} if $x = \frac{1}{2}$. (2)

3.2 Write the following series in sigma notation: $\frac{1}{2} - 2 - \frac{9}{2} - \dots - 37$. (5)

3.3 The sum of the first four terms of a convergent geometric series is 120 and the sum to infinity is $\frac{243}{2}$. Determine the first term of the series. (6)

3.4 The sum of the first n terms of a series is given by $S_n = -3n^2 + 6n$.

3.4.1 Calculate the sum of the first 100 terms. (2)

3.4.2 Determine the 7th term of the series. (3)

[20]

QUESTION 4

Given: $f(x) = 4^{-x}$

- 4.1 Determine the value of $f(-3)$. (1)
- 4.2 Write down the equation of $f^{-1}(x)$ in the form $y = \dots$ (2)
- 4.3 Sketch the graph of f^{-1} showing all relevant information. (3)
- 4.4 Write down the domain of f^{-1} . (1)
- 4.5 For which values of x will $f^{-1}(x) \geq -3$? (2)
- 4.6 If $y = f^{-1}(x+t)$ passes through the point $(68; -3)$, determine the value of t . (1)

[10]

QUESTION 5

Consider the function: $g(x) = \frac{x-5}{x-3}$

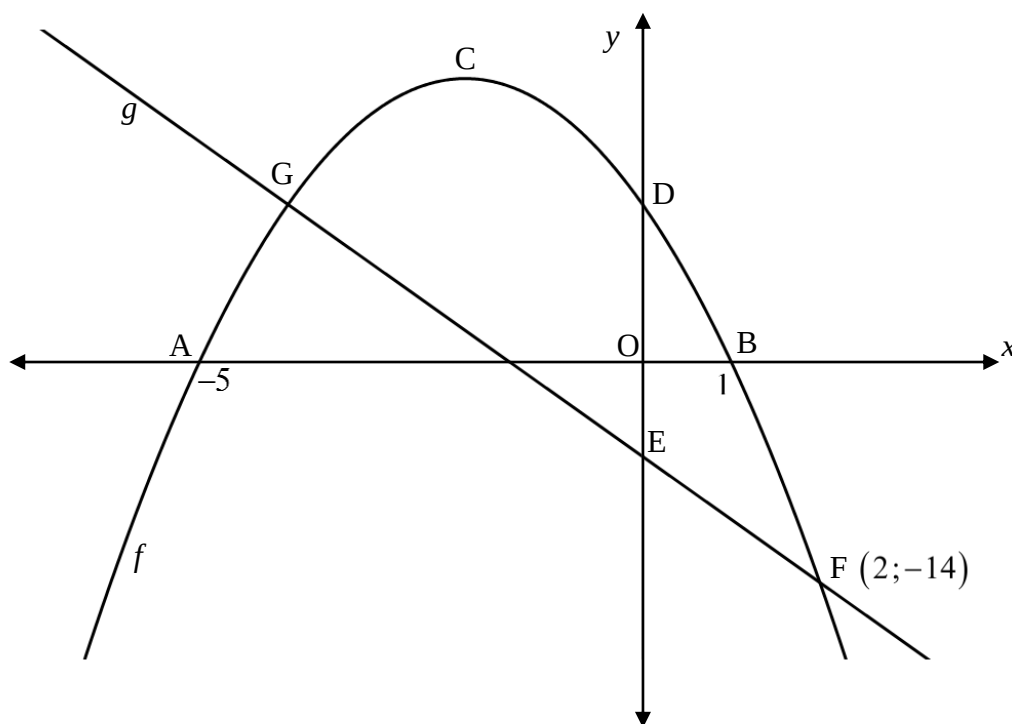
- 5.1 Write down the equation of the vertical asymptote of g . (1)
- 5.2 Write down g in the form $y = \frac{a}{x-p} + q$. (2)
- 5.3 Sketch the graph of g , showing all relevant information. (5)
- 5.4 Determine the equation of the axis of symmetry of g that intersects g . (2)

[10]

QUESTION 6

The graphs of $f(x) = ax^2 + bx + c$ and $g(x) = -4x - 6$ are sketched below.

A $(-5;0)$ and B $(1;0)$ are the x – intercepts of f . C is the turning point of f . D and E are the y – intercepts of f and g respectively. The graphs intersect at F $(2;-14)$ and G.



- 6.1 Show that $a = -2$, $b = -8$ and $c = 10$. (3)
- 6.2 What is the range of f ? (3)
- 6.3 Calculate the length of DE. (2)
- 6.4 For what values of x is $g'(x) \cdot f(x) \leq 0$? (2)
- 6.5 The graph $h(x) = mx + k$ is parallel to g . Determine the value of k for which h is a tangent to f . (5)
- 6.6 Hence, or otherwise, determine the value(s) of k for which $f(x) = h(x)$ has non-real roots. (1)

[16]

QUESTION 7

- 7.1 An asset is purchased for R10 000. It depreciates at a rate of $r\%$ p.a. on the reducing balance. After 3.5 years, the value of the asset is R5 690. Calculate the value of r . (3)
- 7.2 Mr Stippel deposited R9 000 into an account on 1 January 2020. He continues to make monthly deposits of R9 000 at the beginning of every month for 5 years. The account earns interest at 7,5% p.a. compounded monthly. How much money was in the account at the end of 5 years? (5)
- 7.3 A financially wise young professional has accumulated R4,5 million in a retirement fund that will provide a life annuity. He plans to withdraw R50 000 at the end of each month.
- If the fund earns interest at 10% p.a. compounded monthly, calculate the maximum number of withdrawals he will be able to make before the fund is depleted. (6)

[14]

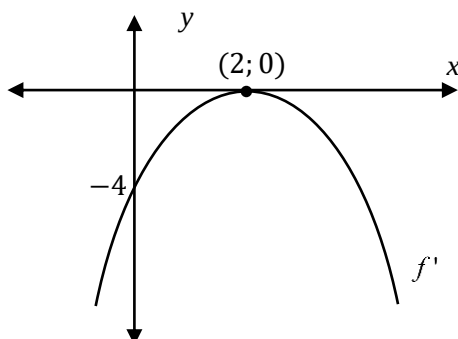
QUESTION 8

- 8.1 If $f(x) = -x^3$, determine $f'(x)$ from first principles. (5)
- 8.2 Determine: $D_x \left[\frac{7}{x} - \sqrt[3]{x} \right]$ (3)
- 8.3 If it is given that $y = kx^2 + k$, determine:
- 8.3.1 $\frac{dy}{dx}$ (1)
- 8.3.2 $\frac{dk}{dy}$ (3)
- 8.4 The curve with equation $y = x - \frac{8}{x}$ passes through the point P (4; q). Determine the gradient of the line perpendicular to the tangent of the curve at P. (3)

[15]

QUESTION 9

The sketch below shows the graph of f' . The turning point of f' is $(2; 0)$ and the y -intercept is $(0; -4)$.

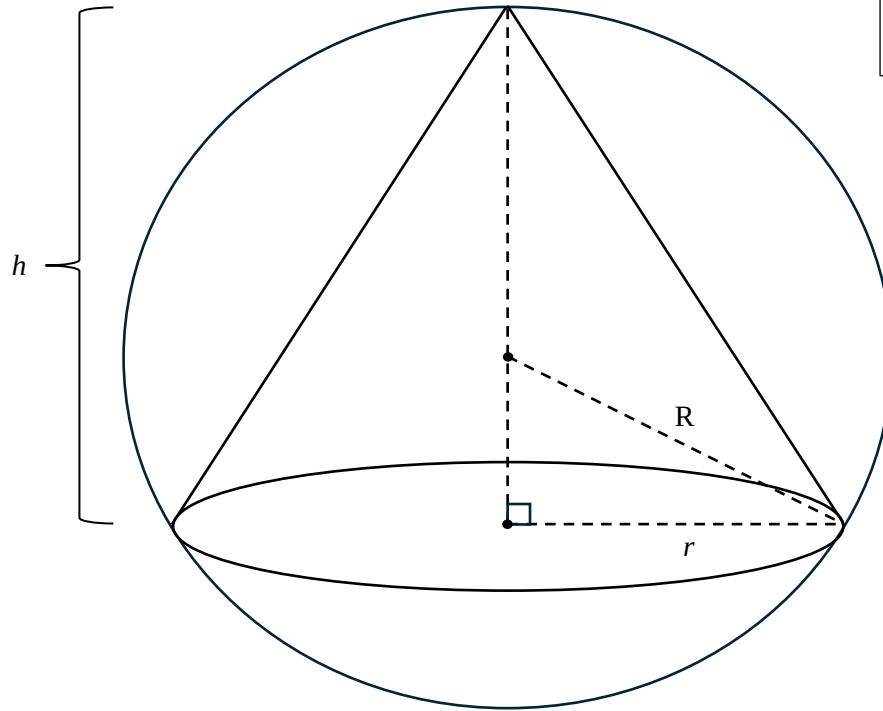


- 9.1 For which value(s) of x is f decreasing? (1)
- 9.2 Write down the x -coordinate of the point of inflection of f . (1)
- 9.3 Is f concave up or concave down at $x = -2$? (1)
- 9.4 Determine the x -values at which the gradient of f is equal to -4 . (2)
- 9.5 Draw a rough sketch of the graph of f if it is given that $f(2) = 9$. (3)

[8]

QUESTION 10

A right circular cone, with a radius of r cm, is inscribed in a sphere with a radius of $R = 6$ cm.



$$\text{Volume of cone} = \frac{1}{3}\pi r^2 h$$

10.1 Show that the volume of the cone can be expressed as $V = \frac{\pi}{3}(12h^2 - h^3)$. (4)

10.2 Calculate the maximum volume of the cone. (5)

[9]

QUESTION 11

A Grade 12 group conducted a survey on learners' study habits and Geography results. Each learner either used flashcards or summaries and either passed or failed Geography. The partially completed table below shows the results of the survey.

Study Method	Passed	Failed
Flashcards	24	6
Summaries	x	2

- 11.1 If 50% of the learners used flashcards, determine the total number of learners in the group. (1)
- 11.2 Hence, determine the value of x . (2)
- 11.3 If a learner was randomly selected from the group, write down the probability that the learner used summaries and failed Geography. (1)
- 11.4 Are the events “using flashcards” and “passing geography” independent? Show calculations to justify your answer. (5)

[9]

QUESTION 12

A school app uses a 4-digit security code. The digits in the code are from 1 to 9. No digit may be repeated.

- 12.1 How many different 4-digit codes can be formed? (2)
- 12.2 How many different 4-digit codes will be even? (2)
- 12.3 Determine the probability that a randomly selected code will be even and the first digit is 4 or 8. (4)

[8]